

PID Control

1. What is a PID Control Loop?

PID stands for:

- **P – Proportional:** Corrects based on how far off you are.
- **I – Integral:** Corrects based on how long you've been off.
- **D – Derivative:** Corrects based on how fast the error is changing.

2. What Is It Used For in Robotics?

- Driving straight
- Turning accurately
- Holding angles or positions
- Line following
- Arm/lift control

3. Sensors Required

- **Gyro Sensor:** Measures angle (used for turning or keeping heading).
- **Built in Gyro Sensor:** Measures angle (used for turning or keeping heading).

4. P Controller Concept (for this class)

We'll use **only P (Proportional)** control for this exercise to keep it simple.

What is error?

In robotics control, **error means the difference between where you *want* to be and where you *are right now*.**

It's a **math value** that tells the robot how far off it is from the goal.

P-only formula:

$\text{error} = \text{target} - \text{current}$

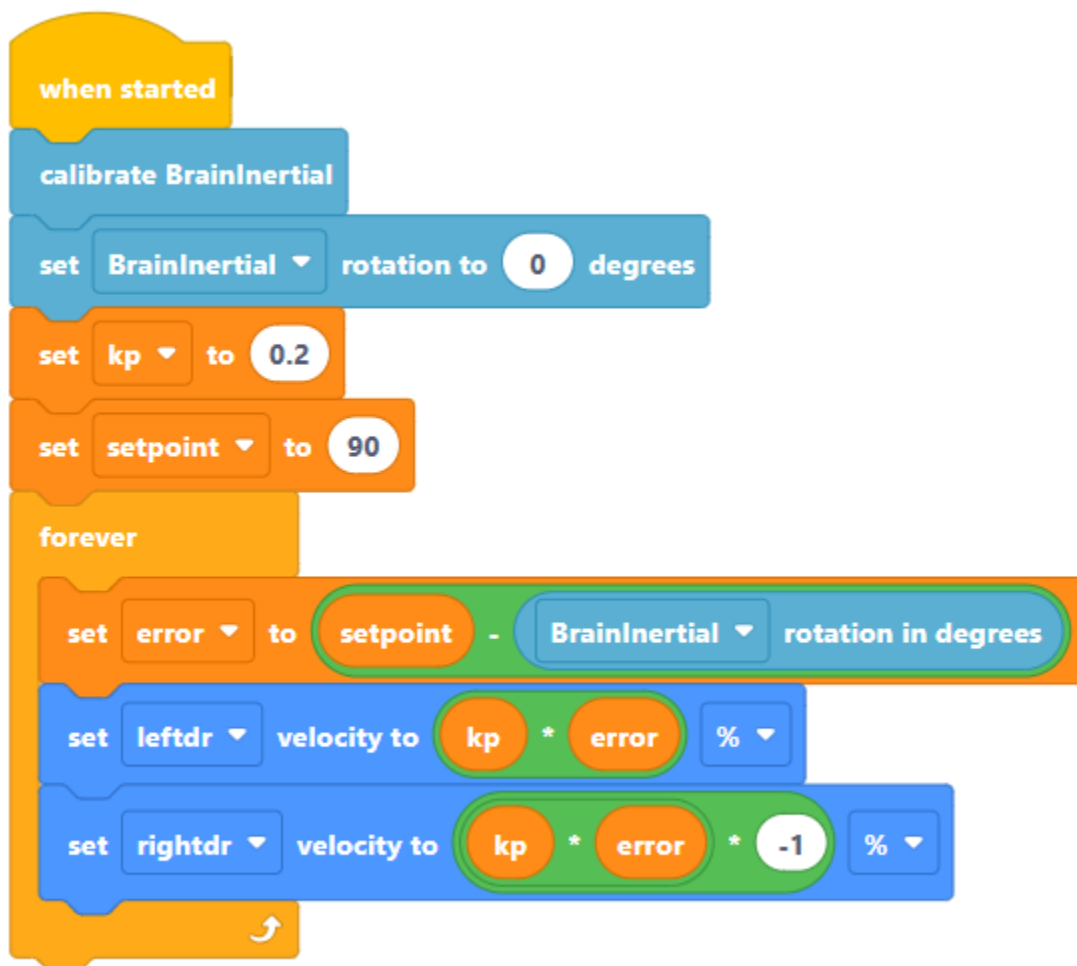
target: The value we want to reach (like angle = 90°, or distance = 1000 mm).

current: The value we are currently at (read from a sensor).

$\text{correction} = kP * \text{error}$

5. Code Structure

Here's a basic outline in (VEXcode IQ):



6. Exercise: Drive in a Square

- Students write a loop to drive forward for X rotations, then turn 90° using gyro + P-control.
- Do this 4 times to complete a square.